

UNIT 2 / FORCES

Quiz 1 quick review • Core ideas + three harder problems from your practice

CORE SKILLS & IDEAS

Start every problem: Draw forces on your chosen object → choose positive directions → write $\Sigma F = ma$ for each axis. Do not add ma as a separate force.

Newton's laws: Zero net force means constant velocity (possibly zero). Net force determines acceleration, not motion direction. Third-law pairs act on **different objects**.

Inclines & components: Gravity contributes $mg \sin \theta$ downhill and $mg \cos \theta$ into the ramp. Resolve every other force separately; a horizontal push has a component into the ramp.

Normal force: Do not automatically use $N = mg$. In an elevator, upward positive gives $N - mg = ma$. Moving downward while slowing means **upward acceleration**.

Connected objects: Find acceleration using the whole system; isolate an object for tension or contact force. One ideal rope has uniform tension; different ropes may not. For an ideal fixed-pulley Atwood machine, $|a| = g|m_2 - m_1|/(m_1 + m_2)$.

Limits & friction: Lift-off begins at $N = 0$; a rope goes slack at $T = 0$. Friction opposes relative slipping or its tendency and can accelerate an object forward.

Forces → motion: For constant acceleration, use $v_f^2 = v_i^2 + 2a\Delta x$ or $\Delta x = v_i t + \frac{1}{2}at^2$. Otherwise, $F_{\text{net}} = m d^2x/dt^2$; integrate F_{net}/m for velocity, then position. A velocity graph's **slope** gives acceleration.

TRY THESE AGAIN

Use $g = 9.80 \text{ m/s}^2$, massless ropes, and ideal pulleys.

1. Horizontal push up a ramp (Practice #3)

A 100 kg crate moves at constant speed up a frictionless 30° ramp under a **horizontal** push. Find the push F and normal force N .

2. Three connected boxes (Practice #11)

Box A (30.0 kg) rests on a frictionless table. A rope runs over an edge pulley to hanging box B (40.0 kg). A second rope joins B to box C (10.0 kg), hanging below it. Released from rest, find both tensions and the distance traveled in 0.250 s.

3. Slack-rope limit (Practice #16)

A 3.0 kg block on a frictionless horizontal surface connects over an edge pulley to a 1.0 kg block on a frictionless 30° downward slope. A horizontal force F pushes the 3.0 kg block **toward the pulley**. Find tension when $F = 2.3 \text{ N}$ and the largest F before the rope goes slack.

QUICK ANSWER CHECK

1. $F = 566 \text{ N}$; $N = 1.13 \times 10^3 \text{ N}$.

$F \cos 30^\circ = mg \sin 30^\circ$; $N = mg \cos 30^\circ + F \sin 30^\circ$.

2. $T_{AB} = 184 \text{ N}$; $T_{BC} = 36.8 \text{ N}$; $d = 0.191 \text{ m}$.

$a = (40 + 10)g/80 = 6.125 \text{ m/s}^2$. The lower rope pulls **downward on B**.

3. $T = 3.10 \text{ N}$; $F_{\text{max}} = 14.7 \text{ N}$.

$F + T = (3.0 \text{ kg})a$; $(1.0 \text{ kg})g \sin 30^\circ - T = (1.0 \text{ kg})a$. Set $T = 0$ for the limit.

Source: your recorded Unit 2 homework and September 26 Newton's Laws Practice, #3, #11, #16.